

Michael Hellstern

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SUMMARY OF QUALIFICATIONS

- **Experience:** Ph.D. statistician with over 7 years of technical experience in the development and application of novel AI and deep learning methodologies and 3 years of industry experience in analytics consulting.
- **Consulting and Collaboration:** Proven ability to bridge the gap between technical methodologies and business value, with specific expertise in managing client relationships and communicating complex concepts to non-expert stakeholders.
- **Technical Lead:** Proven track record of leading end-to-end projects, from problem identification to scalable, high-performance implementation in Python (PyTorch), R, and C++.
- **Technical Skills:** Advanced knowledge of Python and PyTorch for machine learning and GPU acceleration, LangChain, Ollama, vLLM, LLM Foundation Models, SQL, R, C++, SAS.

EDUCATION

- Ph.D. in Biostatistics**, University of Washington Seattle, WA
Thesis: Methods for Time Series Network Analyses (advised by Ali Shojaie) Sep 2020 - Aug 2025
- M.S. in Statistics**, University of Washington Seattle, WA
Jun 2018 - Aug 2020
- B.A. in Economics**, Williams College Williamstown, MA
Sep 2011 - Jun 2015

EXPERIENCE

- University of Washington** Seattle, WA
POSTDOCTORAL RESEARCH Sep 2025 - Present
- Combining LLMs and causal inference to power precision medicine from EHR data. Specifically developing framework to recommend individual treatment plans based on cohort of similar patients.
 - Deployed vLLM on an HPC cluster to serve a 31B language model and built an asynchronous batch processing pipeline to stream and process requests. Successfully processed 100k physician notes in ≈ 11 hours.
- DOCTORAL RESEARCH Jun 2018 - Sep 2025
- Initiated, conceptualized, and developed an end-to-end, model-agnostic, framework to jointly learn AI and deep learning models and detect changes in model performance. Built scalable PyTorch pipeline, including processing of simulation and COVID-19 data. Leveraged GPU acceleration on a HPC.
 - Developed a computationally efficient inference procedure for high-dimensional networks enabling real-time analysis. Used to analyze dynamic brain networks under external stimulation helping treat neurodegenerative disorders.
- STATISTICAL CONSULTING Fall 2022
- Provided guidance on statistical analysis and programming for clients across diverse domains including pediatrics, nutrition sciences, and forestry to translate real-world research questions into statistical and analytical solutions, explained in clear, accessible terms.
- RESEARCH ASSISTANT Jun 2018 - Jun 2019
- Developed and maintain the `netgsa` R package for network-based gene set analysis. Achieved major improvements in computational efficiency - up to 40x faster - through unsupervised clustering and C++ integration.
- Analysis Group** Boston, MA
SENIOR ANALYST Jul 2015 - May 2018
- Collaborated with external clients on analytical projects, translating large-scale insurance claims and EMR data into actionable business insights that resulted in 10 peer-reviewed publications. Conducted analyses in SQL, SAS, and R.
 - Member of experimentation platform team designing core statistical tools and interactive user interface in R and RShiny. Integrated feedback from case teams and clients. Platform was used across dozens of case teams to streamline analysis time from days to hours.

PUBLICATIONS / IN PREPARATION

- [1] **Hellstern M**, Shojaie A. “Dynamic Deep Learning for Detecting Changes in Biomedical Processes”. *Under review*.
- [2] **Hellstern M**, Shojaie A. “Order Selection in Vector Autoregression by Mean Square Information Criteria”. DOI: [10.48550/arXiv.2511.19761](https://doi.org/10.48550/arXiv.2511.19761)
- [3] **Hellstern M**, Kim B, Shojaie A. “Assumption-Lean Inference for Spectral Differential Network Analysis for High-dimensional Time Series”. *In preparation*. *Draft available on request*.
 - * Received a *Best Student Paper Award* from the ASA Statistical Learning and Data Science (SLDS) section. Presented at JSM 2025.
- [4] **Hellstern M**, Kim B, Harchaoui Z, Shojaie A. “Spectral Differential Network Analysis for High-dimensional Time Series”. *AISTATS (2025)*. DOI: [10.48550/arXiv.2412.07905](https://doi.org/10.48550/arXiv.2412.07905)
 - * Received a *Best Student Poster Honorable Mention* from the ASA Statistical Learning and Data Science (SLDS) Section.
- [5] **Hellstern M**, Ma J, Yue K, Shojaie A. “netgsa: Fast computation and interactive visualization for topology-based pathway enrichment analysis”. *PLOS Computational Biology* 17.6 (2021): e1008979. DOI: [10.1371/journal.pcbi.1008979](https://doi.org/10.1371/journal.pcbi.1008979)
- [6] Desai U, Kirson NY, Kim J, Khunti KK, King SB, Trieschman E, **Hellstern M**, Hunt PR, Mukherjee J. “Time to Treatment Intensification After Monotherapy Failure and Its Association With Subsequent Glycemic Control Among 93,515 Patients With Type 2 Diabetes”. *Diabetes care* 41.10 (2018): 2096-2104. DOI: [10.2337/dc17-0662](https://doi.org/10.2337/dc17-0662)
- [7] DerSarkissian M, Lefebvre P, Joshi K, Brown B, Lafeuille MH, Bhak R, **Hellstern M**, Bobbili P, Shiner B, El Khoury A, Young-Xu Y. “Health Care Resource Utilization and Costs Associated with Transitioning to 3-month Paliperidone Palmitate Among US Veterans”. *Clinical Therapeutics* 40.9 (2018): 1496-1508. DOI: [10.1016/j.clinthera.2018.07.011](https://doi.org/10.1016/j.clinthera.2018.07.011)
- [8] Fuld A, Young-Xu Y, Li S, Pilon D, Bhak R, Duchesneau E, **Hellstern M**, Kamstra R, Behl A, Lefebvre P, Freedland S. “Predictors of overall survival (OS) in veterans with non-metastatic castration resistant prostate cancer (nmCRPC)”. *Journal of Clinical Oncology* 36 (2018): e17057-e17057. DOI: [10.1200/JCO.2018.36.15_suppl.e17057](https://doi.org/10.1200/JCO.2018.36.15_suppl.e17057)

HONORS & AWARDS

- *Best Student Paper Award* from the ASA Statistical Learning and Data Science (SLDS) 2025
- *Best Student Poster Honorable Mention* from the ASA SLDS 2024
- *Donovan J. Thompson Award* for best combined performance on PhD qualifying exams in UW Biostatistics 2021

INVITED TALKS

- Assumption-Lean Inference for Spectral Differential Network Analysis of High-Dimensional Time Series**
 - Joint Statistical Meetings (JSM) 2025
 - Western North American Region of the International Biometric Society (WNAR) 2025
- Spectral Differential Network Analysis for High-Dimensional Time Series**
 - Western North American Region of the International Biometric Society (WNAR) 2024
 - Joint Conference on Computational and Financial Econometrics and Computational and Methodological Statistics (CFE-CMStatistics) 2024

SERVICE

- Reviewer for Journal of Machine Learning Research, Journal of the Royal Statistical Society: Series B, Journal of the American Statistical Association

INTERESTS

- Mountain biking, whitewater kayaking, trail running, skiing, ski touring